CREATING THE FUTURE

YOUR WAY

VIETTEL HIGH TECHNOLOGY

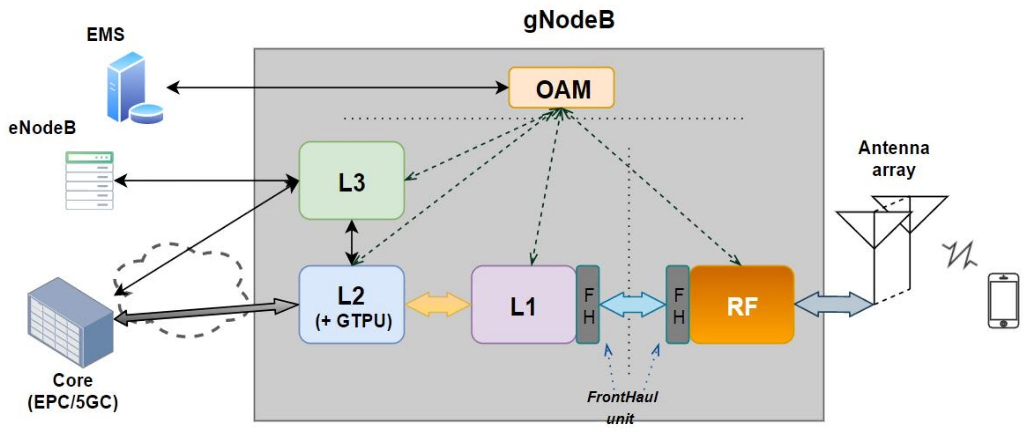
VIETTEL 5G TALENT 2024

5G RAN OVERVIEW

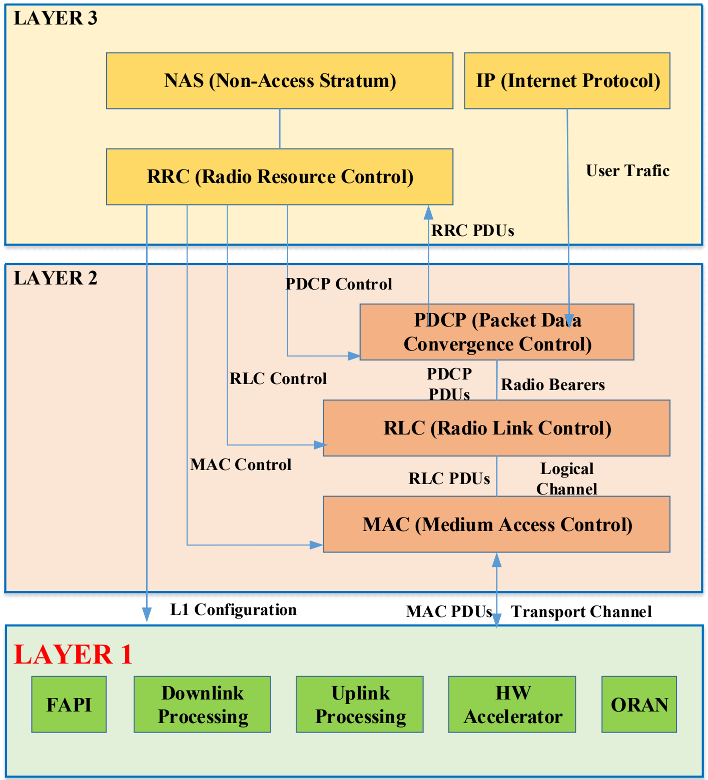
Hà Nội 05/2024

5G GNODEB ARCHITECTURE & SW PROTOCOL STRUCTURE

WHERE IS LAYER 2/3?

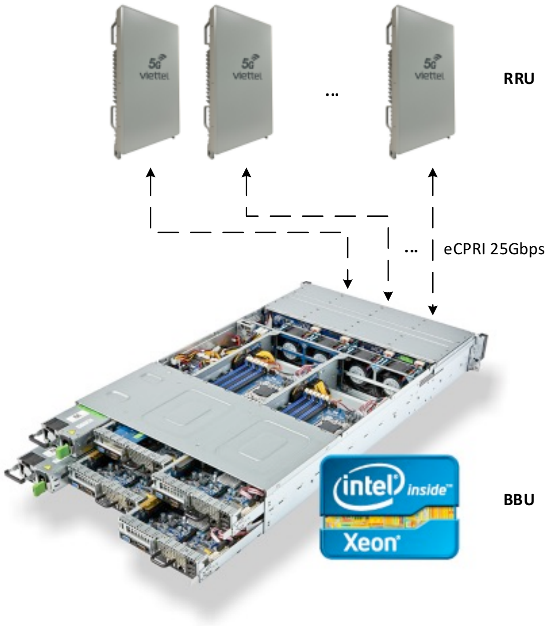


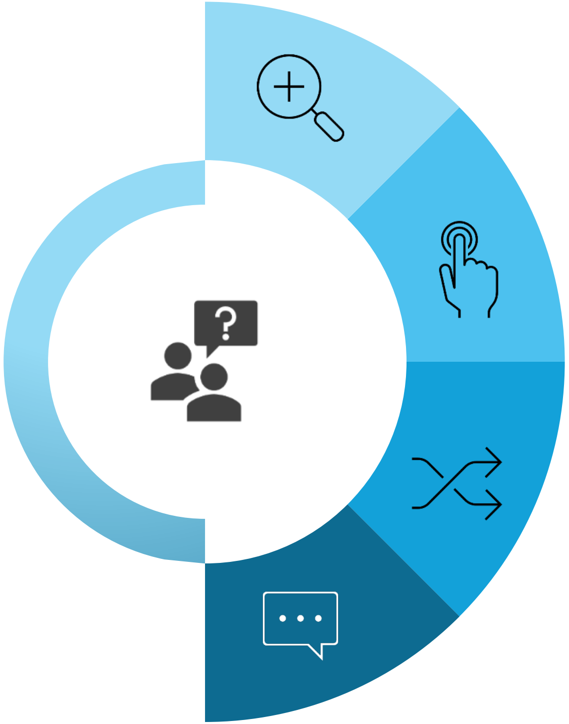
GNODEB ARCHITECTURE



GNODEB SOFTWARE PROTOCOL

5G LAYER 2-3 JOB DESCRIPTION





PROJECT

4G/5G

HW PLATFORM

Intel
Qualcom, NXP

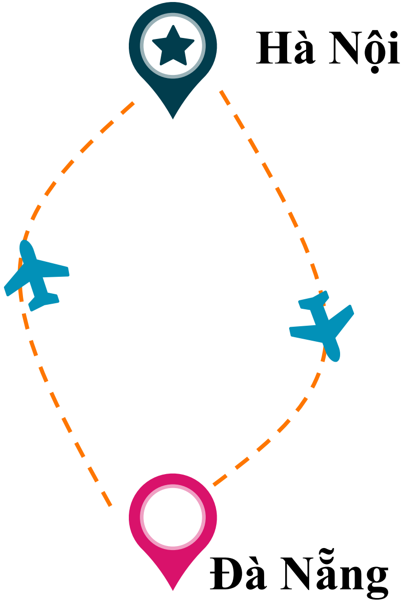
WORK ITEM

Research, Development
Integration, testing, debugging
Optimization

ACKNOWLEDGE & SKILL

Network protocols
C, Linux Kernel
Test equipments

5G R&D PROTOCOL (LAYER 2+3) TEAM



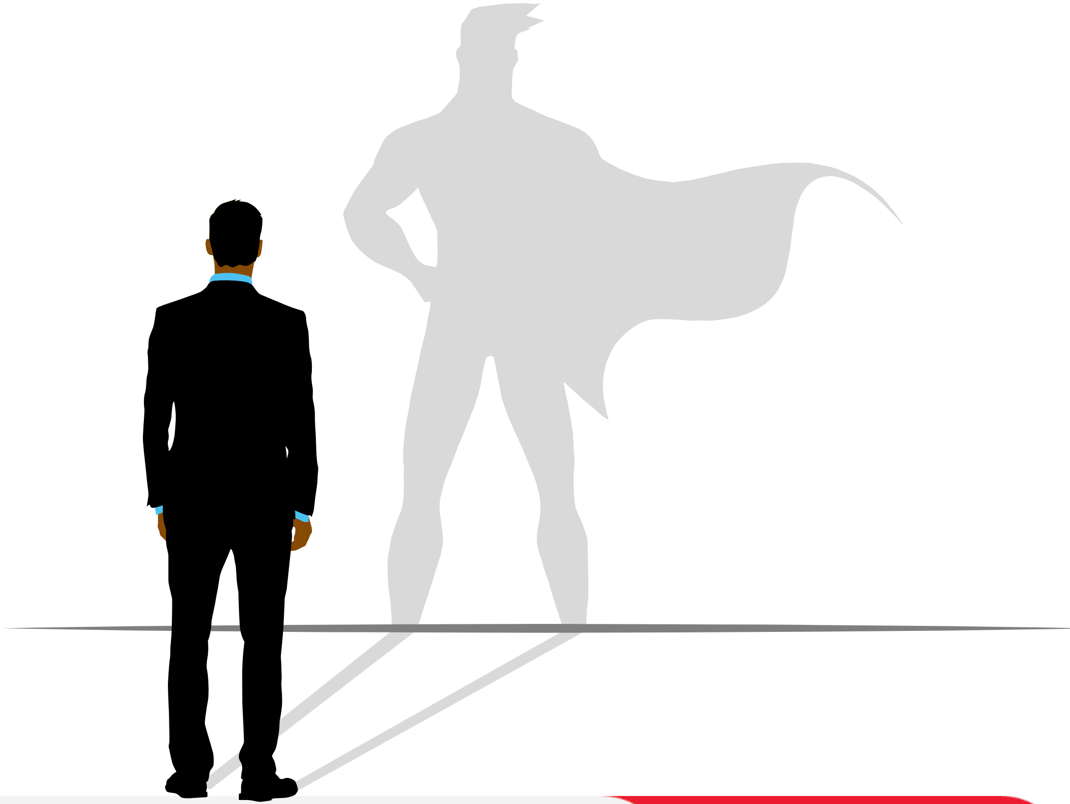
5G LAYER 2+3 EMPLOYEE TRAINING

DURATION

- 3 months – 6 months

TRAINING TOPIC

- C language
- Linux OS, Linux programming
- 5G Layer 3 protocols
- 5G Layer 2 protocols
- Techniques for Accelerating 5G throughput
- Test and measurement equipment
- SW integration
- System integration



Summary

I. 5G Review

II. 5G Protocol Stack

III. Paging call flow and feature

IV. 5G NR SA Registration / Attach call flow (Not Study)

V. Mini-project

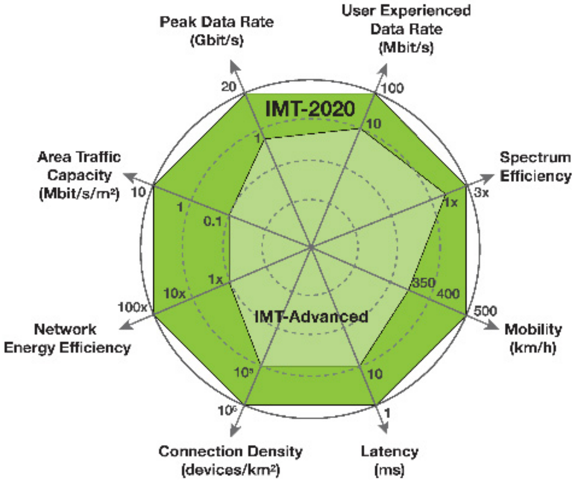
I. Review

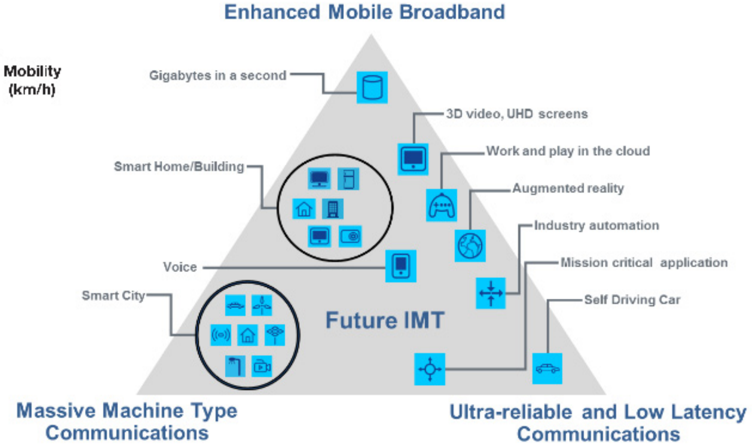
I. 5G Review

1. 5G
2. ITU
3. 3GPP
4. Network Architecture Overview
5. 4G/5G Non-standalone Architecture (NSA)
6. 5G Standalone Architecture (SA)
7. 5G Throughput

1. 5G

- Object
  - EMBB
  - URLLC
  - Mass type connections
- Network Elements
  - UE,
  - Radio Access Network
  - Transport Network
  - Core Network
- Key Technology ?
  - Massive MIMO
  - Beamforming
  - Network slicing
  - Network flexibility – Virtualization





2. ITU – International Telecommunication Union

- ITU
    - Founded 1985
    - Agency of United Nation (UN)
    - 193 member states + 900 companies, universities and organization
    - Coordinates the global use of the radio spectrum and defines requirements for next-generation mobile communication technologies.
- 3 main sectors
    - ITU-R Radio Communication
    - ITU-T Standardization (called Recommendation)
    - ITU-D Development

Committed to connecting the world

- Output document
    - IMT-2000: define “3G”; year of release: 2000
    - IMT-Advanced: define “4G”; year of release: 2010
    - IMT-2020: define “5G”; year of release 2020

3. 3GPP – 3rd Generation Partnership Project

- 3GPP
  - 1998
  - Is an standard organization
  - From 7 regional organization partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC) + 700 companies
  - Create & maintain technical standard for global mobile communication technologies as GSM, UMTS, LTE, NR

3GPP™

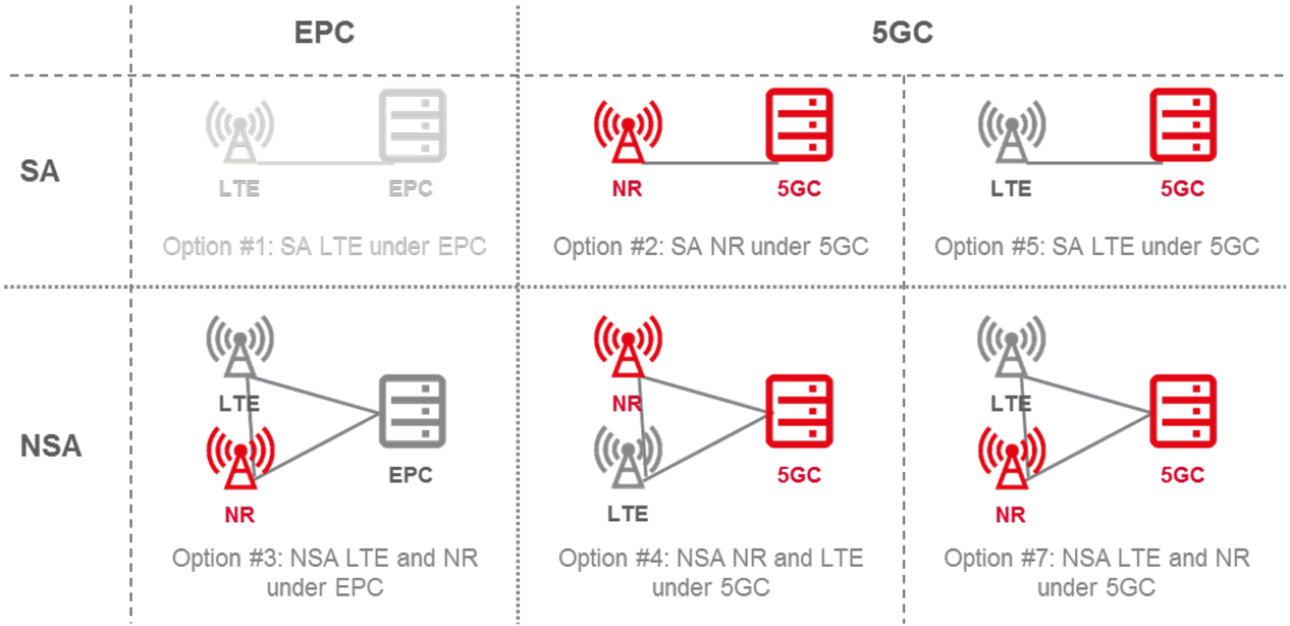
A GLOBAL INITIATIVE

- 3 main group
  - Radio Access Network
  - Service & System Aspects
  - Core Network & Terminal

- Output document
  - Release 99: “3G” UMTS standard
  - Release 8: “first 4G” LTE standard; year of release: 2008
  - Release 15: “first 5G”; NR standard; year of release: 2018

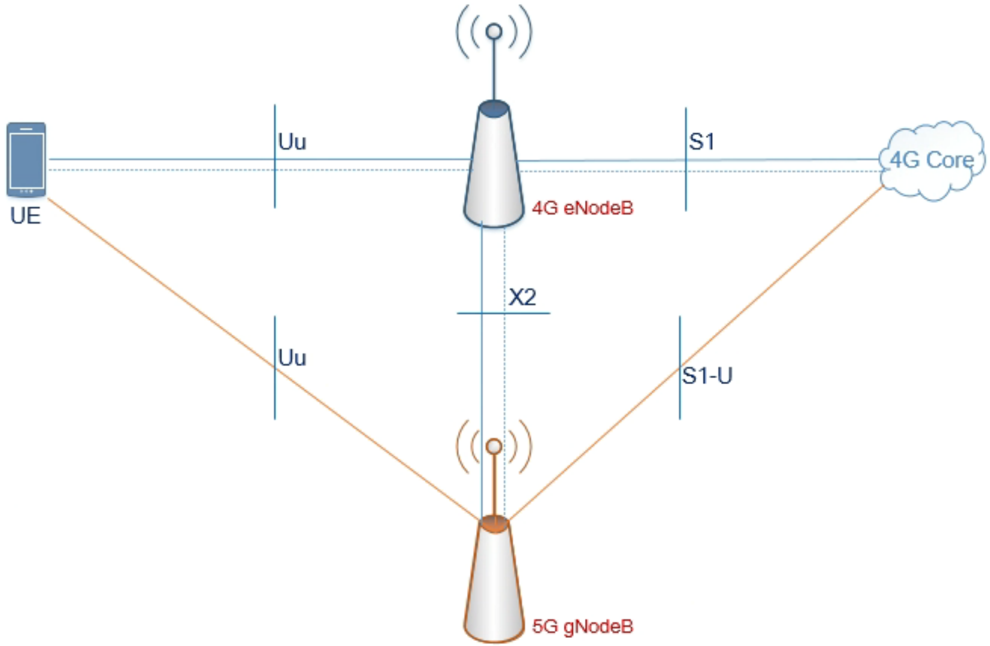
4. 4G/5G Network Architecture Overview

- 4G LTE option 1
- 4G/5G NSA option 3
- 5G SA option 2



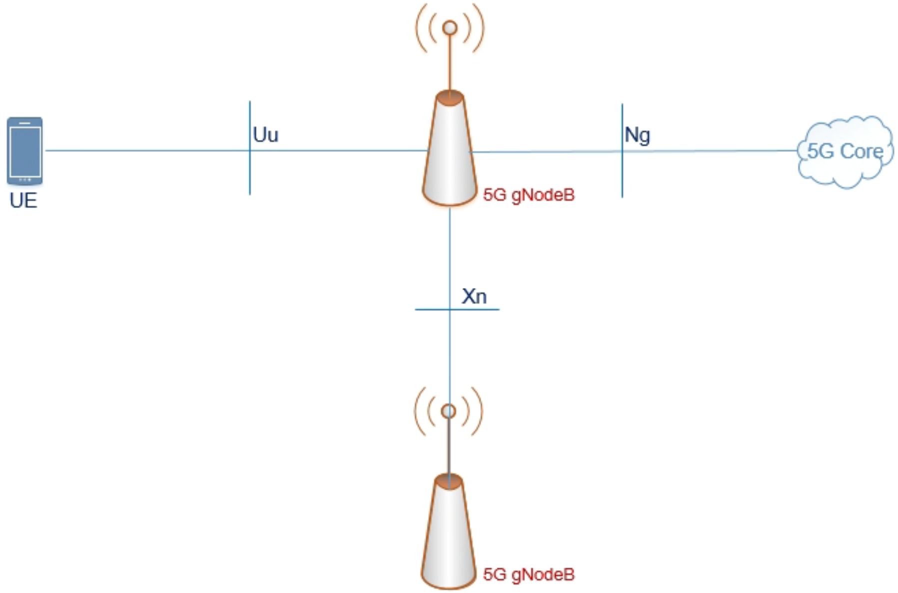
5. Non-standalone Architecture (NSA)

- NSA option 3
- Radio access network is composed of
  - 4G eNodeB as master node
  - 5G gNodeB as secondary node
- Core is 4G core
- Advance
  - Quickly deploy
  - Add only 5G gNodeB
  - Increase throughput
- Dis-advance
  - Adapt only MBB
  - Complex operation



6. 5G Standalone Architecture (SA)

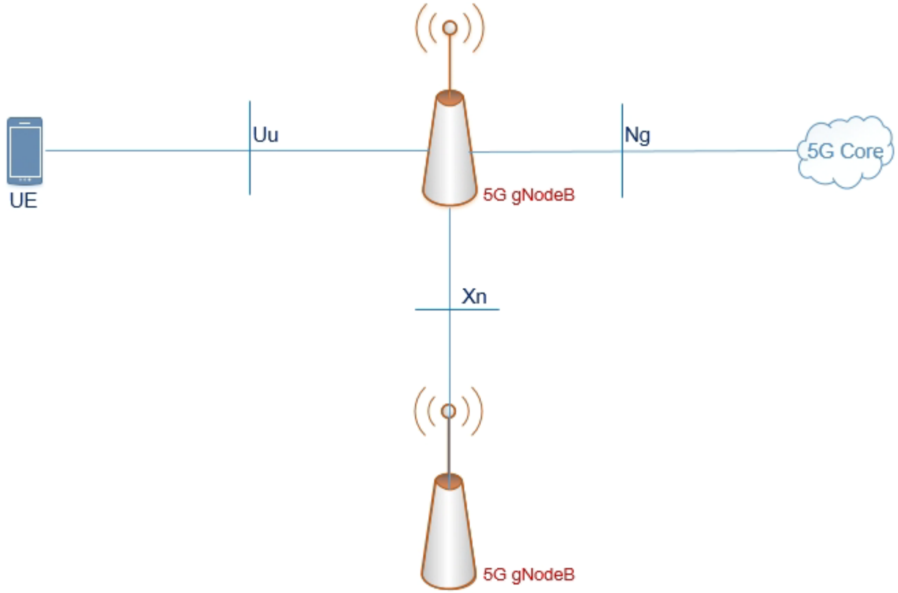
- SA option 2
- Radio Access Network
  - 5G gNodeB
- Core Network:
  - 5G Core
- Advance
  - Full 5G technology to achieve: MBB, ULLC, Mass types of connection
  - Simple operation:



6. Student work

- Why 5G throughput is higher than 4G throughput ?

- Calculation the peak 5G throughput

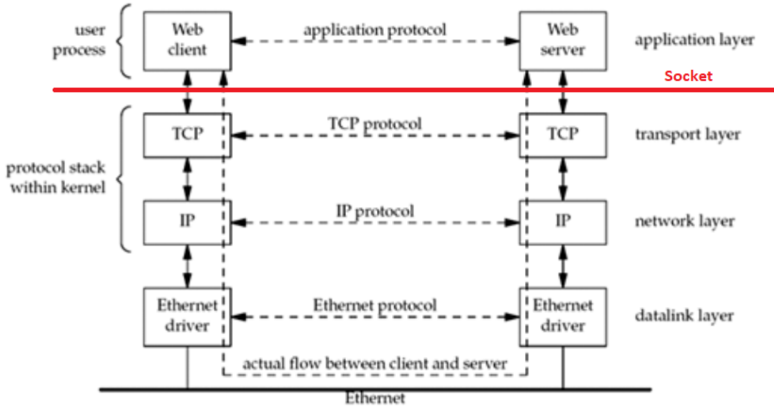


II. 5G Protocol Stacks – overview

II. 5G Protocol Stack - gNodeB

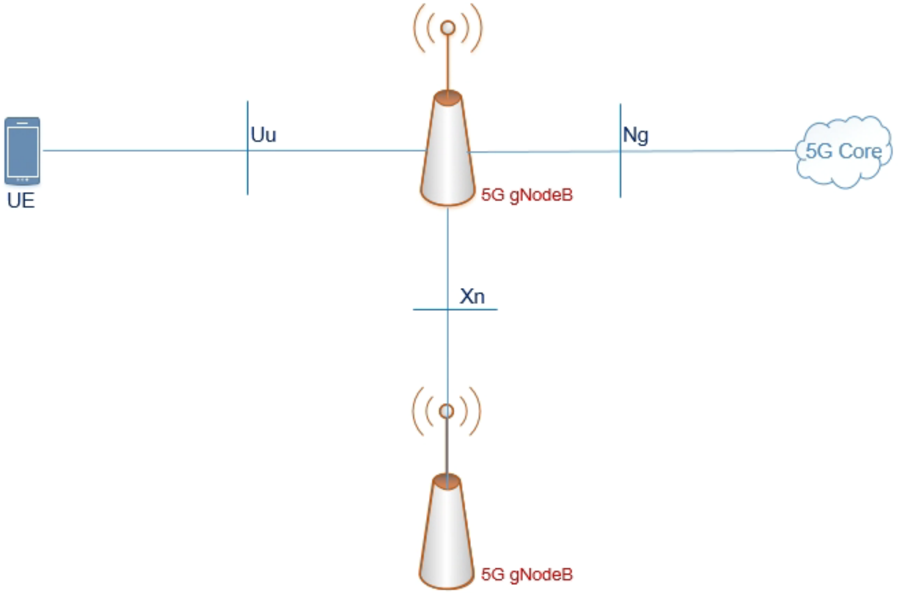
1. Definition

- Protocol
  - A set of rules or procedures for transmitting data between electronic devices
  - Protocols in Transport layer: ?
- Interface
  - A shared boundary across which two or more components
- Protocol Stack
  - A set of protocols used in communication network --- ?

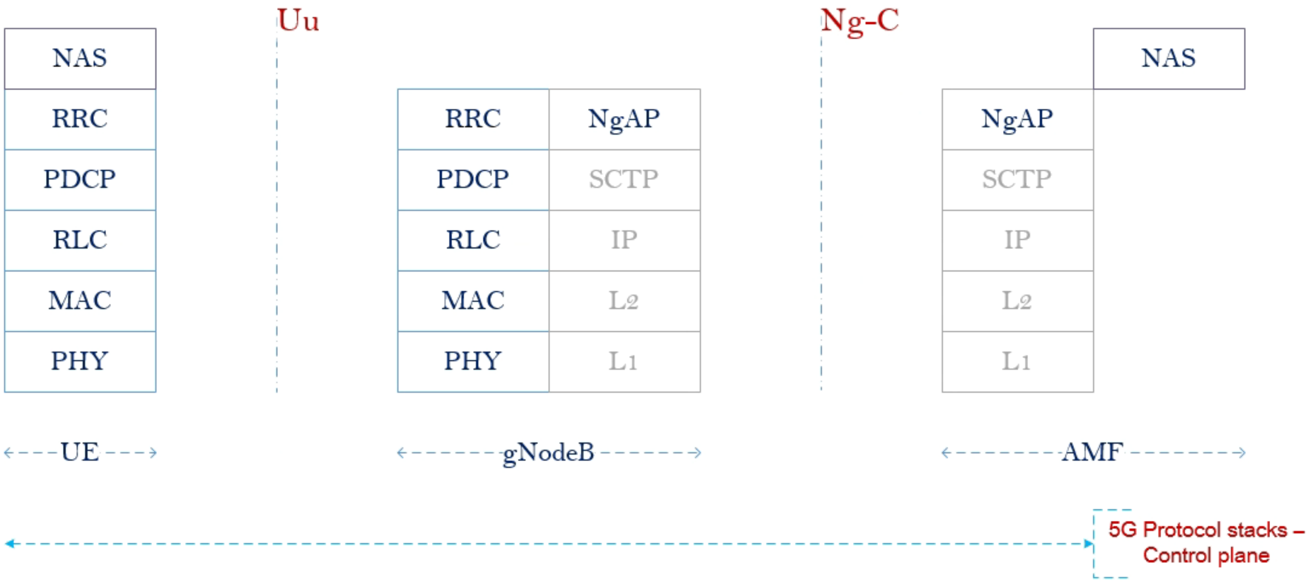


2. gNodeB Interfaces

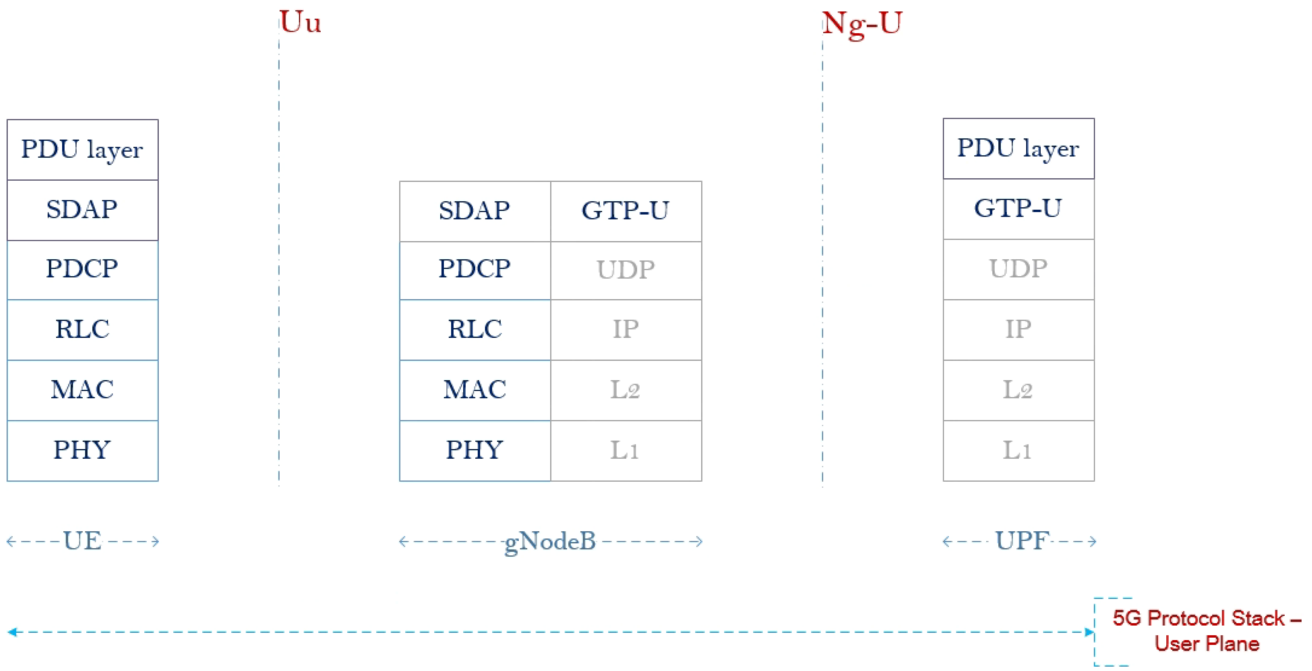
- Uu Interface or Radio Interface
  - interface between UE and 5G gNodeB
- Ng Interface or N2 Interface
  - Interface between gNodeB and 5G Core network
    - Ng-C interface for control plane
    - Ng-U interface for user plane
  - Ref: from TS 38.401 to TS 38.414
- Xn Interface
  - Interface between gNodeb and gNodeB
    - Xn-C interface for control plane
    - Xn-U interface for user plane
  - Ref: from TS 38.420 to TS 38.424



3. gNodeB Protocol Stack - Control plane



4. gNodeB Protocol Stack - User plane



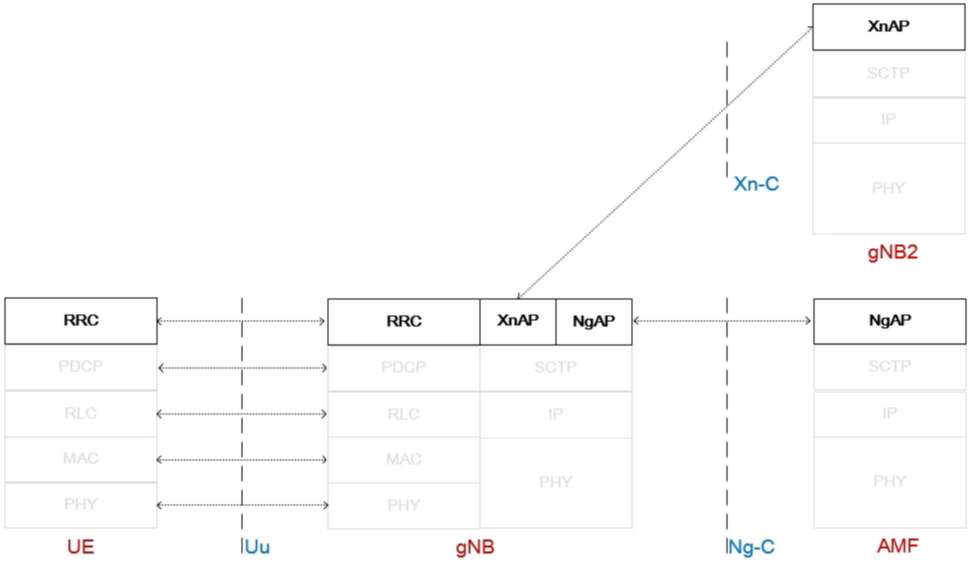
4. Student work

Show the gNodeB Protocol Stack for Xn Interface – Control Plane & User Plane

5 minutes

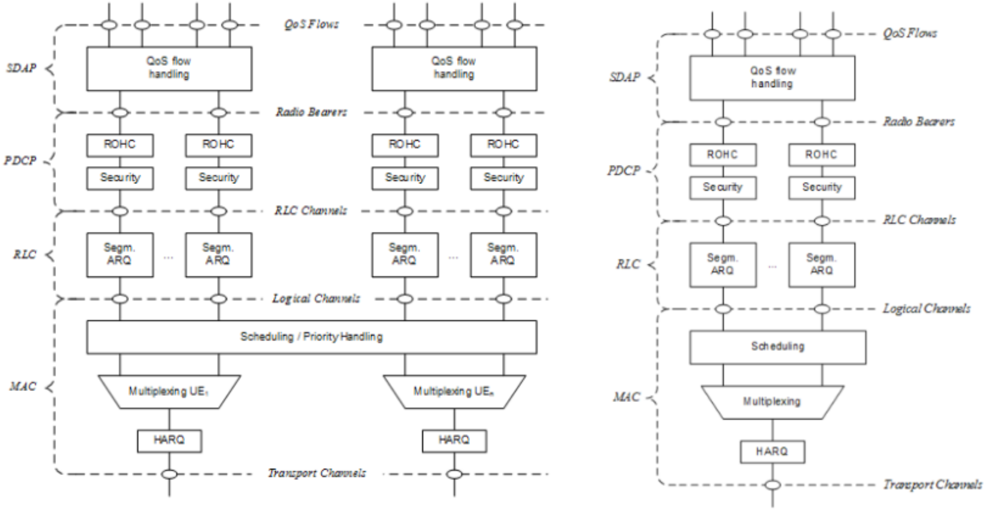
5. 5G NR Layer 3

- Protocol
  - Radio Resource Control - RRC
  - Ng Application Protocol – NgAP
  - Xn Application Protocol – XnAP
  - F1 Application Protocol – F1AP
- Main function
  - Broadcast of system information
  - Manage RRC connection (establish, maintain, release)
  - Manage Data Radio Bearer – DRB (establish, configuration, maintain, release)
  - Mobility function & QoS management
  - Manage the procedures between gNB and 5GC
  - Manage the procedures among gNBs



6. 5G NR Layer 2

- Service Data Application Protocol - SDA
- Package Data Convengence Protocol – PDCP
    - Sequence numbering
    - Ciphering & integrity
    - Header compression
- Radio Link Control – RLC
    - 3 mode of transfer: UM, AM, TM
    - Segmentation and re-segmentation
    - Error correction through ARQ
- Medium Access Control – MAC
    - Mapping logical-transport channels
    - Multiplexing/ de-multiplexing.
    - Scheduling
    - Error correction through HARQ



7. 5G NR Layer 1

Design to meet the main 5G use cases

- eMBB
  - More Bandwidth/ subcarrier
  - Dynamic TDD
  - Massive MIMO/Beam forming
  - More OFDM symbol per slot
- URLLC
  - Shorter slot
  - Less OFDM symbol per slot
- mMTC
  - Flexible slot
  - Flexible subcarrier spacing

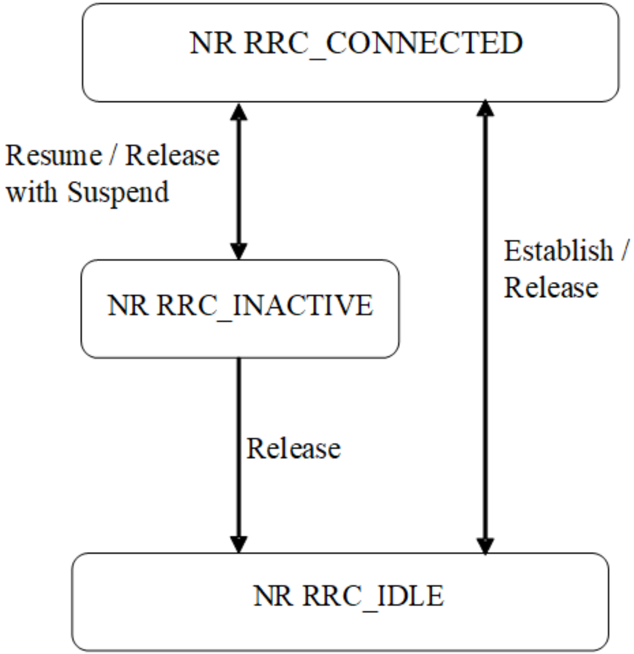
	LTE	5G-NR
Maximum Bandwidth (per CC)	20 MHz	50 MHz (@ 15 kHz), 100 MHz (@ 30 kHz), 200 MHz (@ 60 kHz), 400 MHz (@ 120 kHz)
Subcarrier Spacing	15 kHz	$2^n \cdot 15$ kHz TDM and FDM multiplexing
Waveform	CP-OFDM for DL; SC-FDMA for UL	CP-OFDM for DL; CP-OFDM and DFT-s-OFDM for UL
Maximum Number of Subcarriers	1200	3300
Subframe Length	1 ms (moving to 0.5 ms)	1 ms
Latency (Air Interface)	0.5 ms	0.5 ms
Slot Length	7 symbols in 500 µs	14 symbols (duration depends on subcarrier spacing) 2, 4 and 7 symbols for mini-slots
Channel Coding[2]	Turbo Code (data); TBCC (control)	Polar Codes (control); LDPC (data)
Initial Access	PVS[1] Beamforming in Digital Domain	Beamforming
MIMO	8x8	8x8
Reference Signals	UE Specific DMRS and Cell Specific RS	Front-loaded DMRS (UE-specific)
Duplexing	FDD, Semi-Static TDD, Half-Duplex FDD, Dynamic TDD	FDD, Semi-Static TDD, Half-Duplex FDD, Dynamic TDD

1. Precoding Vector Switching
2. LTE: Turbo coding for PDSCH/PUSCH, TBCC for PDCCH/PBCH, simplex coding for PHICH, RM/dual-RM for PUCCH
   NR: Polar coding for PDCCH UL UCI (DL; CA Polar, UL; PC Polar), LDPC for PDSCH/PUSCH, RM for small size of data for PUCCH

Comparing L1 of LTE against 5G. Source: Intel 2018, slide 14.*

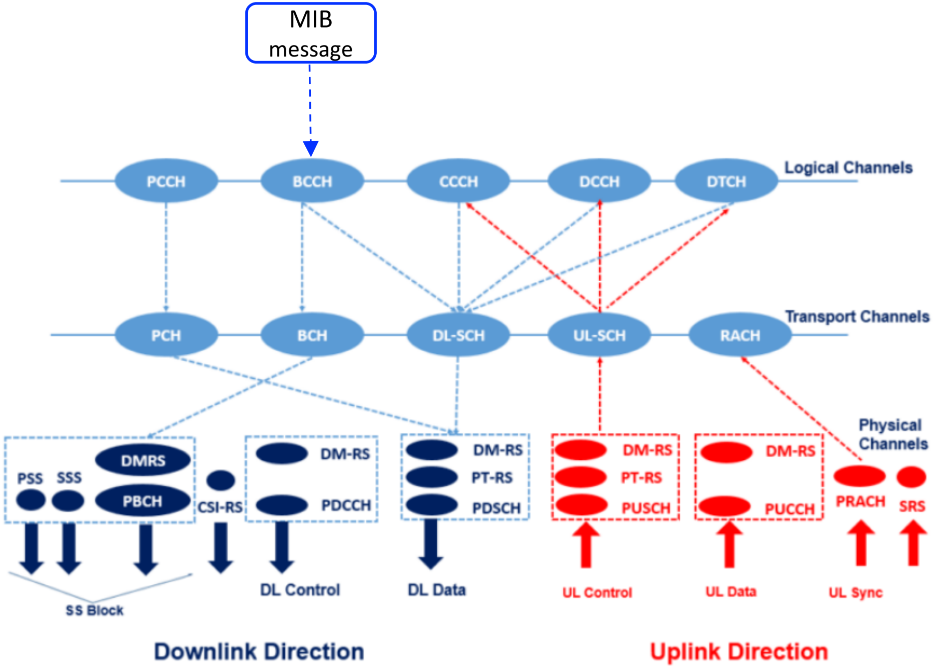
8. 5G NR Layer 3: RRC Protocol

- RRC: Radio Resource Control (RRC) protocol is used in on the Air Interface.
  - 5GPP TS 38.331
- Procedures: (by 5. Procedures in TS.38.331)
- RRC states in 5G New Radio
  - NR-RRC Connected
  - NR-RRC Inactive
  - NR-RRC Idle
- Functions:
  - Broadcast of system information
  - RRC connection control
  - Bearer management
  - Inter-RAT mobility
  - Measurement configuration and reporting



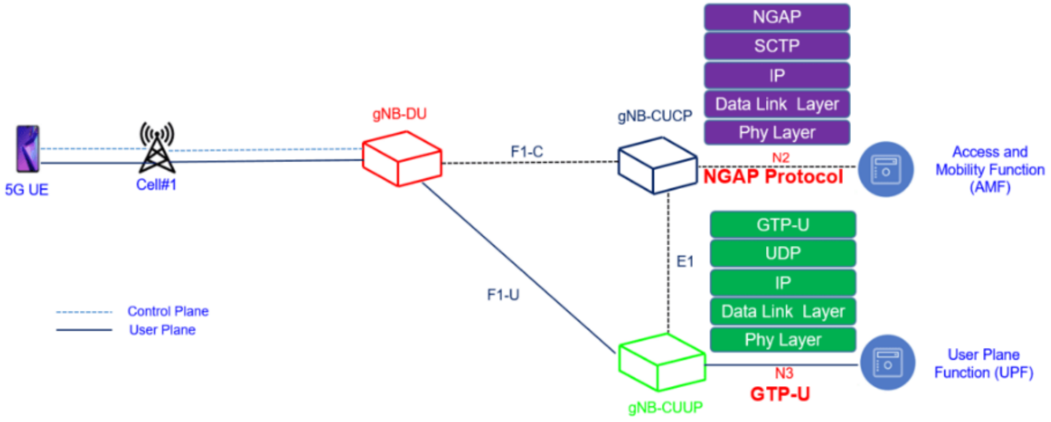
10. 5G NR Layer 3: RRC Protocol

- MIB (RRC messages) - ~ 24bit
    - RLC mode: TM
    - Logical channel: BCCH
    - Transport channel: BCH
    - Physical channel: PBCH
        - 576 Resource Elements (240*2 + 48*2)
        - Coding: polar coding
        - Modulation: QPSK
        - antenna transmission scheme: single



9. 5G NR Layer 3: NgAP protocol

- Types of Services over NGAP
  - Non UE-associated services
  - UE-associated services
- Types of NgAP procedure
  - Request Response Procedure
  - No Response Procedure
- Reference
  - 4GPP TS 38.413 5G NG-RA Protocol (NGAP)



9. 5G NR Layer 3: NgAP prococol – Request Response procedures

Elementary Procedure	Initiating Message	Successful Outcome	Unsuccessful Outcome
		Response message	Response message
AMF Configuration Update	AMF CONFIGURATION UPDATE	AMF CONFIGURATION UPDATE ACKNOWLEDGE	AMF CONFIGURATION UPDATE FAILURE
RAN Configuration Update	RAN CONFIGURATION UPDATE	RAN CONFIGURATION UPDATE ACKNOWLEDGE	RAN CONFIGURATION UPDATE FAILURE
Handover Cancellation	HANDOVER CANCEL	HANDOVER CANCEL ACKNOWLEDGE	
Handover Preparation	HANDOVER REQUIRED	HANDOVER COMMAND	HANDOVER PREPARATION FAILURE
Handover Resource Allocation	HANDOVER REQUEST	HANDOVER REQUEST ACKNOWLEDGE	HANDOVER FAILURE
Initial Context Setup	INITIAL CONTEXT SETUP REQUEST	INITIAL CONTEXT SETUP RESPONSE	INITIAL CONTEXT SETUP FAILURE
NG Reset	NG RESET	NG RESET ACKNOWLEDGE	
NG Setup	NG SETUP REQUEST	NG SETUP RESPONSE	NG SETUP FAILURE
Path Switch Request	PATH SWITCH REQUEST	PATH SWITCH REQUEST ACKNOWLEDGE	PATH SWITCH REQUEST FAILURE
PDU Session Resource Modify	PDU SESSION RESOURCE MODIFY REQUEST	PDU SESSION RESOURCE MODIFY RESPONSE	
PDU Session Resource Modify Indication	PDU SESSION RESOURCE MODIFY INDICATION	PDU SESSION RESOURCE MODIFY CONFIRM	
PDU Session Resource Release	PDU SESSION RESOURCE RELEASE COMMAND	PDU SESSION RESOURCE RELEASE RESPONSE	
PDU Session Resource Setup	PDU SESSION RESOURCE SETUP REQUEST	PDU SESSION RESOURCE SETUP RESPONSE	
UE Context Modification	UE CONTEXT MODIFICATION REQUEST	UE CONTEXT MODIFICATION RESPONSE	UE CONTEXT MODIFICATION FAILURE
UE Context Release	UE CONTEXT RELEASE COMMAND	UE CONTEXT RELEASE COMPLETE	
Write-Replace Warning	WRITE-REPLACE WARNING REQUEST	WRITE-REPLACE WARNING RESPONSE	
PWS Cancel	PWS CANCEL REQUEST	PWS CANCEL RESPONSE	
UE Radio Capability Check	UE RADIO CAPABILITY CHECK REQUEST	UE RADIO CAPABILITY CHECK RESPONSE	

9. 5G NR Layer 3: NgAP prococol – No Response procedures

Elementary Procedure	Message
Downlink RAN Configuration Transfer	DOWNLINK RAN CONFIGURATION TRANSFER
Downlink RAN Status Transfer	DOWNLINK RAN STATUS TRANSFER
Downlink NAS Transport	DOWNLINK NAS TRANSPORT
Error Indication	ERROR INDICATION
Uplink RAN Configuration Transfer	UPLINK RAN CONFIGURATION TRANSFER
Uplink RAN Status Transfer	UPLINK RAN STATUS TRANSFER
Handover Notification	HANDOVER NOTIFY
Initial UE Message	INITIAL UE MESSAGE
NAS Non Delivery Indication	NAS NON DELIVERY INDICATION
Paging	PAGING
PDU Session Resource Notify	PDU SESSION RESOURCE NOTIFY
Reroute NAS Request	REROUTE NAS REQUEST
UE Context Release Request	UE CONTEXT RELEASE REQUEST
Uplink NAS Transport	UPLINK NAS TRANSPORT
AMF Status Indication	AMF STATUS INDICATION
PWS Restart Indication	PWS RESTART INDICATION
PWS Failure Indication	PWS FAILURE INDICATION
Downlink UE Associated NRPPa Transport	DOWNLINK UE ASSOCIATED NRPPA TRANSPORT
Uplink UE Associated NRPPa Transport	UPLINK UE ASSOCIATED NRPPA TRANSPORT
Downlink Non UE Associated NRPPa Transport	DOWNLINK NON UE ASSOCIATED NRPPA TRANSPORT
Uplink Non UE Associated NRPPa Transport	UPLINK NON UE ASSOCIATED NRPPA TRANSPORT
Trace Start	TRACE START
Trace Failure Indication	TRACE FAILURE INDICATION
Deactivate Trace	DEACTIVATE TRACE
Cell Traffic Trace	CELL TRAFFIC TRACE
Location Reporting Control	LOCATION REPORTING CONTROL

III. 5G NR Paging Feature

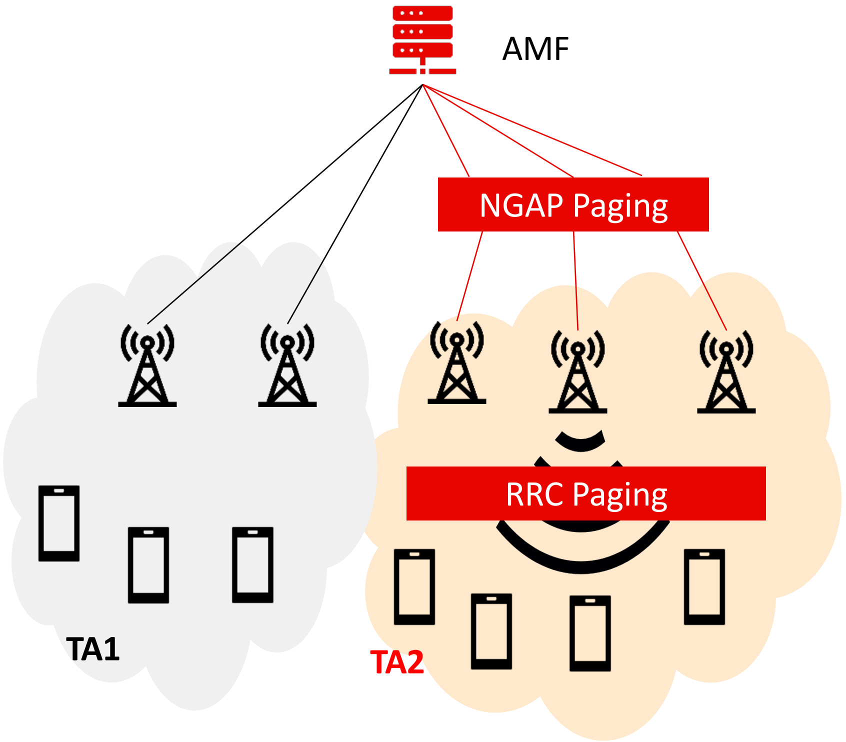
III. 5G NR Paging Feature

1. What - Requirement
2. Paging Call Flow
3. 3GPP – document
4. Paging Occasion
5. Calculate the Paging Occasion at UE
6. Select the best Paging Occasion at gNB
7. Simulation
8. Mini-project

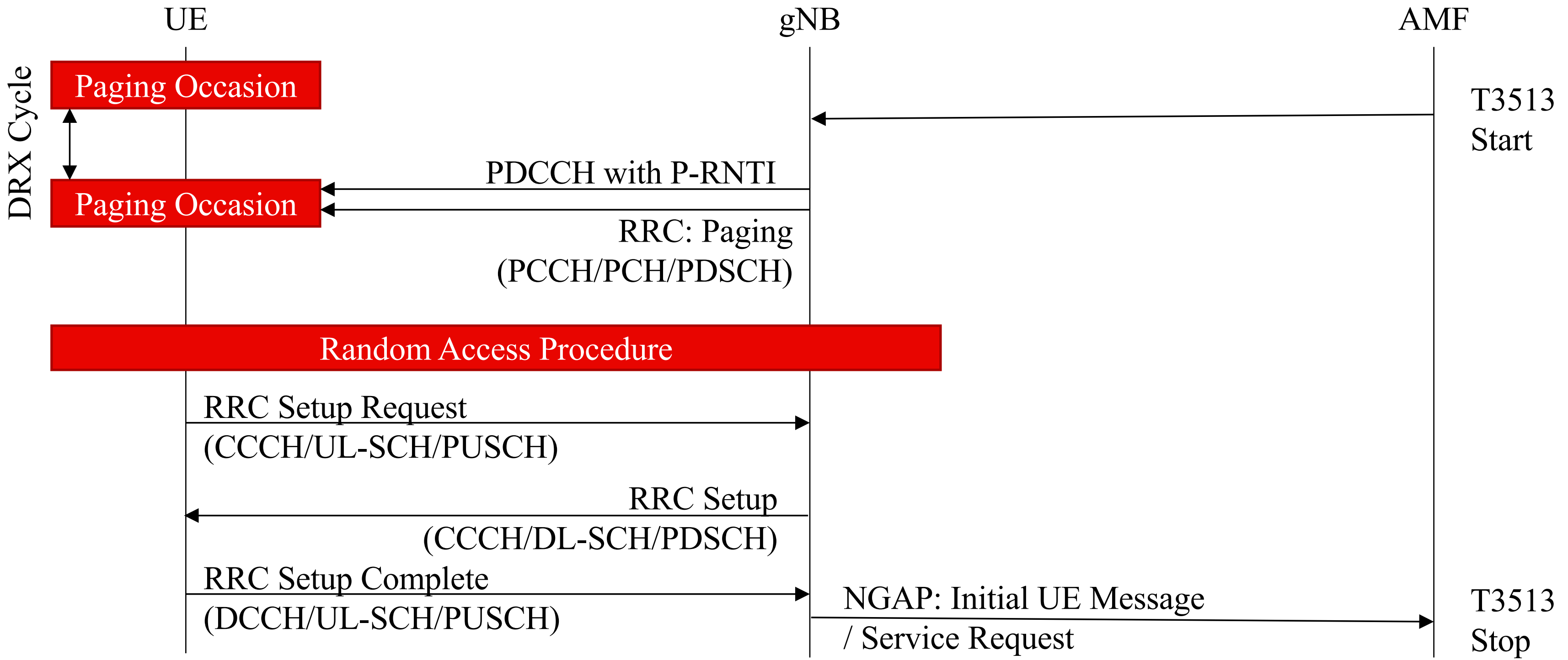
1. What - Requirement

Purpose of Paging Procedure:

- Triggering mobile terminated connections setup (RRC Setup Request/ RRC Resume Request)
- Informing UE of the System Information Modification
- Notifying UE of incoming ETWS/CMAS message



2. Paging Call Flow



3. 3GPP – Paging document

TS 38.331	5G; NR; Radio Resource Control (RRC); Protocol specification	RRC Paging Procedure
TS 38.413	5G; NG-RAN; NG Application Protocol (NGAP)	NGAP Paging Procedure
TS 38.423	5G; NG-RAN; Xn Application Protocol (XnAP)	XnAP Paging Procedure
TS 38.304	5G; NR; User Equipment (UE) procedures in idle mode and in RRC Inactive state	Discontinuous Reception for paging
TS 38.213	5G; NR; Physical layer procedures for control	PDCCH Paging Search Space

3. 3GPP – NgAP Paging message

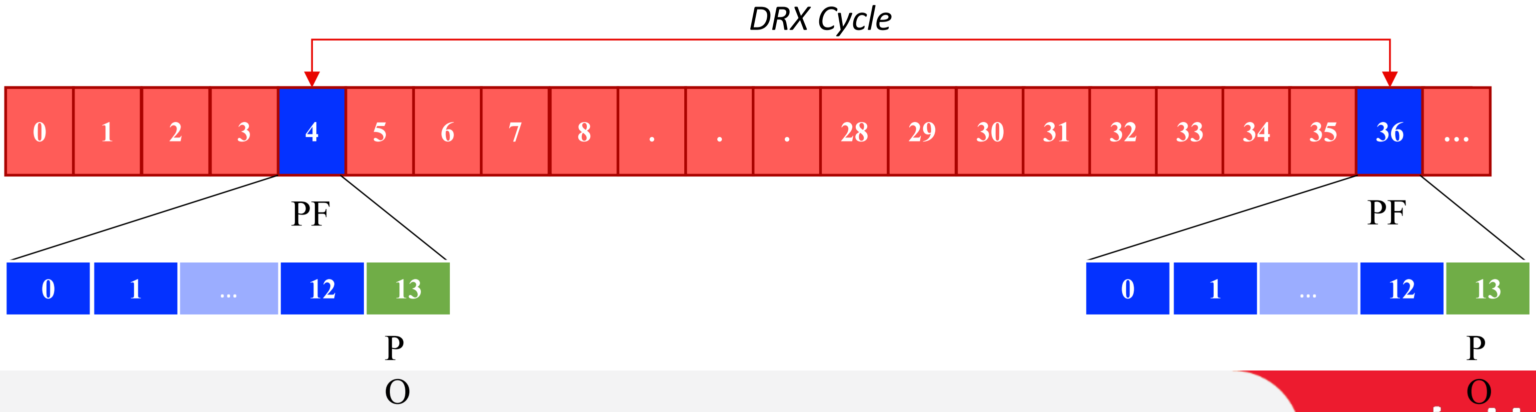
<table>
<tr><th colspan="5">NGAP: Paging <TS 38.413></th></tr>
<tr><td>UE Paging Identity (Mandatory)</td><td colspan="2">5G-S-TMSI</td><td colspan="2">AMF Set ID & AMF Pointer & 5G-TMSI</td></tr>
<tr><td>Paging DRX</td><td colspan="4">32, 64, 128, 256</td></tr>
<tr><td rowspan="2">TAI List for Paging (Mandatory)</td><td colspan="4">SEQUENCE {1 to 16 instances}</td></tr>
<tr><td colspan="2">Tracking Area ID</td><td colspan="2">PLMN Id & Tracking Area Code (TAC)</td></tr>
<tr><td>Paging Priority</td><td colspan="4">1, 2, 3, 4, 5, 6, 7, 8</td></tr>
<tr><td rowspan="4">UE Radio Capability for Paging</td><td rowspan="2">Capability for NR</td><td rowspan="2" colspan="2">List of Supported NR Bands</td><td>SEQUENCE {1 to 1024 instance}</td></tr>
<tr><td>1-1024</td></tr>
<tr><td rowspan="2">Capability for E-UTRA</td><td rowspan="2" colspan="2">List of Supported E-UTRA Bands</td><td>SEQUENCE {1 to 64 instance}</td></tr>
<tr><td>1-256</td></tr>
<tr><td rowspan="2">Assistance Data for Paging</td><td colspan="4">Recommended Cells</td></tr>
<tr><td colspan="4">Paging Attempt Info</td></tr>
<tr><td>Paging Origin</td><td colspan="4">Non-3GPP</td></tr>
</table>

3. 3GPP – RRC Paging message

RRC Paging <3GPP TS 38.331>				
Paging Record List	SEQUENCE {1 to 32 instances}			
	PagingRecord	ue-Identity	CHOICE	
			ng-5G-S-TMSI	Full I-RNTI
			BIT STRING {48 bit}	BIT STRING {40 bit}
		accessType	non3GPP	

4. Paging Occasion

- Paging Occasion (PO)
  - A Subframe when UE switch to active mode to check for Paging message
- Paging Frame (PF)
  - A Radio Frame may contain one or multiple PO(s)
- DRX Cycle
  - Period between two Paging Occasion (UE sleep mode duration)



5. Calculate the Paging Occasion at UE

According to 3GPP TS 38.304:

- Paging Frame:

  (SFN + PF_offset) mod T = (T div N)*(UE_ID mod N)

- Paging Occasion:

  i_s = floor (UE_ID/N) mod Ns

  - T: DRX cycle of the UE
  - N: Number of total paging frames in T
  - Ns: number of paging occasions for a PF
  - PF_offset: offset used for PF determination
  - UE_ID: 5G-S-TMSI mod 1024

  Note: All Parameters except for UE_ID are signaled in ***SIB1***.

6. Example: calculation the PF/PO at UE

SIB1

```
pcch-Config
{
  defaultPagingCycle rf128,
  nAndPagingFrameOffset oneT : NULL,
  ns one,
  firstPDCCH-MonitoringOccasionOfPO :
    {
      2
    }
}
```

log_ue_paging.txt

Paging

```
2021 Aug  3  08:56:49.805  [CA]  0xB821  NR5G RRC OTA Packet  --  PCCH / Paging
Subscription ID = 1
Misc ID         = 0
Pkt Version = 9
RRC Release Number.Major.minor = 15.9.0
Radio Bearer ID = 0, Physical Cell ID = 200
Freq = 515790
Sfn = 128, SubFrameNum = 1
slot = 0
PDU Number = PCCH Message,    Msg Length = 10
SIB Mask in SI =  0x00

Interpreted PDU:

value PCCH-Message ::=
{
  message c1 : paging :
      {
        pagingRecordList
        {
          {
            ue-Identity ng-5G-S-TMSI : '00000000 01000001 11010000 00000000 00010110 01000001'B
          }
        }
      }
}
```

PF/PO of this UE ???

7. Example- Select the best Paging Occasion at gNB

- gNodeB receives the NgAP Paging message
  - At current gNB_SFN : 100 // [0-1023]
  - 5G-S-TMSI:

```
ue-Identity ng-5G-S-TMSI : '00000000 01000001 11010000 00000000 00010110 01000001'B
```

PF/PO that gNodeB Should be sent to this UE
???

THANK YOU

VIETTEL HIGH TECHNOLOGY

VIETTEL 5G TALENT 2023

5G RAN PART 3
LAYER 3: ARCHITECTURE, FUNCTIONS & PROCEDURES

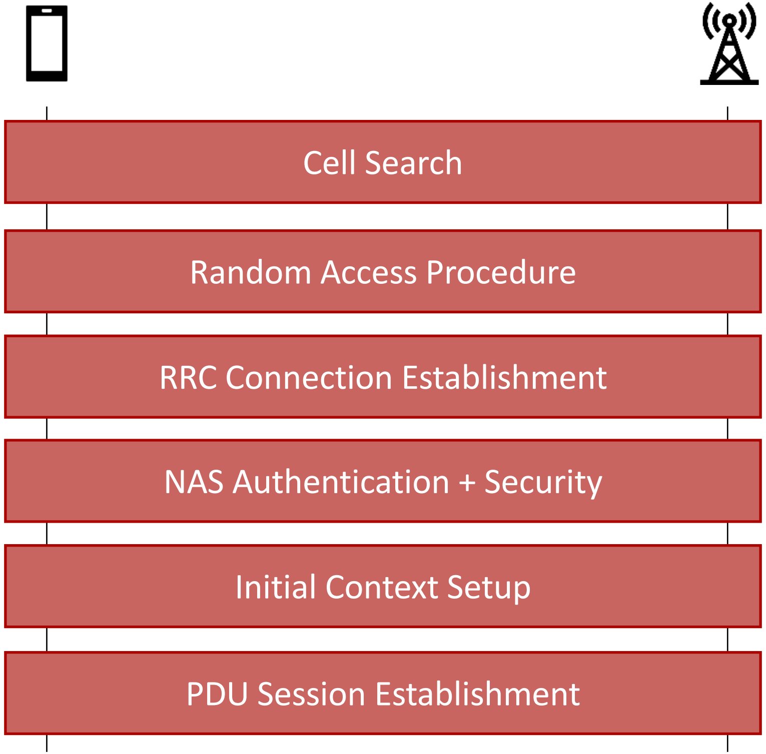
Hà Nội 04/2023

IV. 5G NR SA Registration/Attach

IV. 5G NR SA Registration/Attach

1. Registration Call Flow
2. Cell Search
3. Random Access Procedure
4. RRC Connection Establishment
5. Initial Context Setup
6. PDU Session Establishment
7. Simulation
8. Mini-project

1. Registration Call Flow



2. Cell Search

Definition <3GPP TS 38.300>:

- Cell search is the procedure by which a UE acquires time and frequency synchronization with a cell and detects the physical Cell ID of that cell.

2. Cell Search

Frequency Tuning:

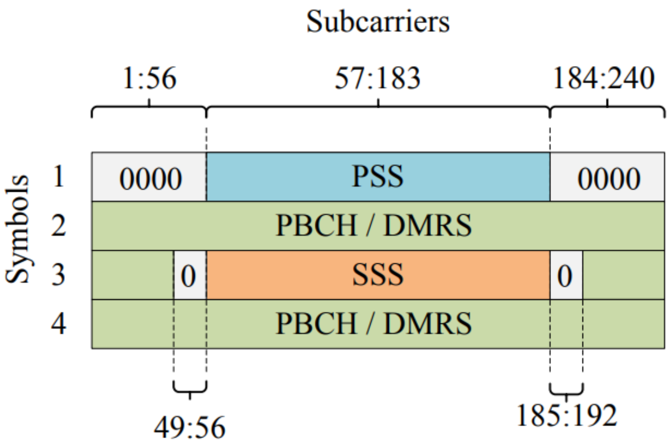
- UE detects Synchronization Signal Block based on:
    - Band Scan (RSSI Scan)
    - Synchronization Raster (GSCN) Scan

Frequency range	SS Block frequency position SS_{REF}	GSCN	Range of GSCN
0 – 3000 MHz	N * 1200 kHz + M * 50 kHz, N=1:2499, M ϵ {1,3,5} (Note 1)	3N + (M-3)/2	2 – 7498
3000 – 24250 MHz	3000 MHz + N * 1.44 MHz N = 0:14756	7499 + N	7499 – 22255
24250 – 100000 MHz	24250.08 MHz + N * 17.28 MHz, N = 0:4383	22256 + N	22256 – 26639
NOTE 1: The default value for operating bands with which only support SCS spaced channel raster(s) is M=3.			

2. Cell Search

Synchronization Signal Block (SSB):

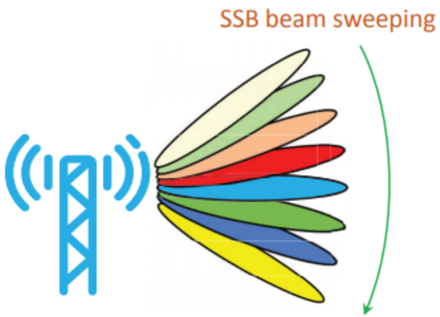
- Primary Synchronization Signal (PSS)
    - Psuedo-random m-sequence
    - Determine $N^{(2)}_{Cell\ Id} = \{0, 1, 2\}$
- Secondary Synchronization Signal (SSS)
    - Psuedo-random Gold sequence
    - Determine $N^{(1)}_{Cell\ Id} = \{0, 1, \dots, 335\}$
    - $N_{Cell\ ID} = 3N^{(1)}_{Cell\ Id} + N^{(2)}_{Cell\ Id}$
- PBCH DMRS
    - Reference Signal for PBCH Demodulation
- PBCH
    - Decode Master Information Block

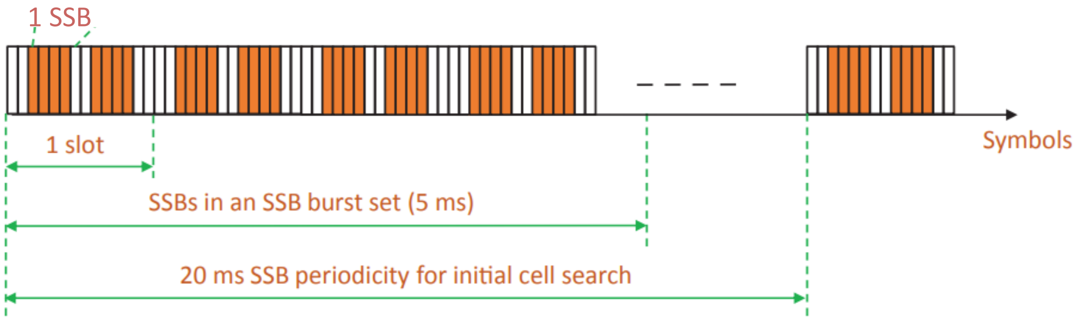


2. Cell Search

SSB – Time Domain:

- Transmit in a SSB Burst
    - Include maximum L SSB within a half frame (5 ms)
        - Value of L based on SCS and frequency band
    - Each SSB (identified by SSB index) is allocated to a beam
- SSB periodicity = {5, 10, 20, 40, 80, 160} ms





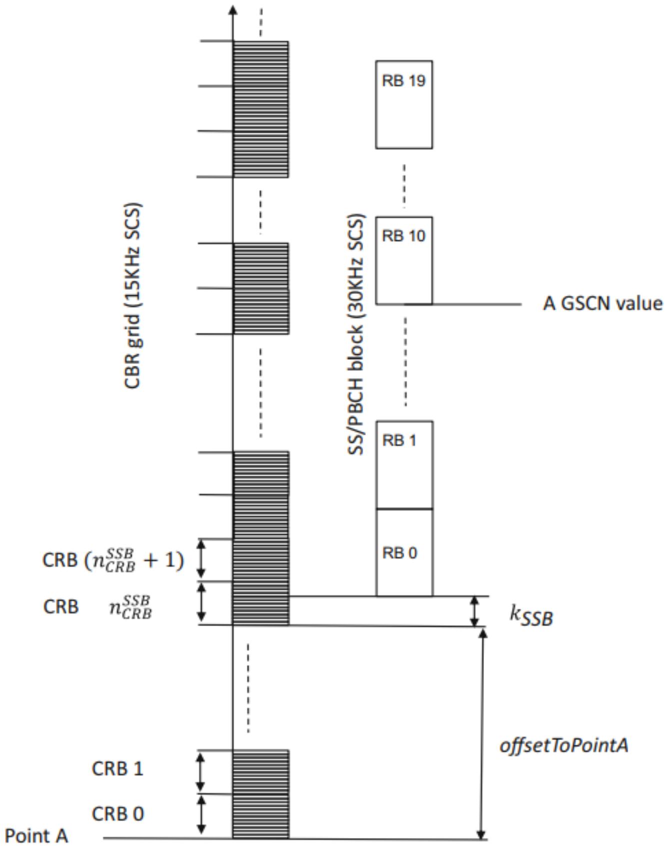
2. Cell Search

SSB Position in SSB Burst <3GPP TS 38.213>				
Subcarrier Spacing	1st symbol position	Frequency Band		
		< 3 GHz	> 3 GHz, ≤ 6 GHz	> 6 GHz
Case A: 15 kHz	{2,8} + 14 n	n = 0,1 (Lmax = 4)	n = 0,1,2,3 (Lmax = 8)	-
Case B: 30 kHz	{4,8,16,20}+28n	n = 0 (Lmax = 4)	n = 0,1 (Lmax = 8)	-
Case C: 30 kHz	{2,8} + 14 n	(f < 3) n = 0,1	n = 0,1,2,3 (FDD)	-
		(f < 1.88) n = 0,1	n = 0,1,2,3 (TDD)	
		(1.88 < f < 3.0) n = 0,1,2,3	n = 0,1,2,3 (TDD)	
Case D: 120 kHz	{4,8,16,20} + 28n			n=0, 1, 2, 3, 5, 6, 7, 8, 10, 11, 12, 13, 15, 16, 17, 18 (Lmax = 64)
Case E: 240 kHz	{8, 12, 16, 20, 32, 36, 40, 44} + 56n			n=0, 1, 2, 3, 5, 6, 7, 8 (Lmax = 64)

2. Cell Search

SSB – Frequency Domain:

- Initial Blind Search can only determined GSCN value
- The position of SSB relative to Point A can be determined by MIB and SIB1 information
    - offsetToPointA
    - ssb-SubcarrierOffset (kSSB)



2. Cell Search

- After PBCH/MIB decoding, UE can achieve radio frame, slot and symbol synchronization and complete the Cell Search procedure
  - systemFrameNumber:
    - 6 MSB of 10 bit SFN in MIB
    - 4 LSB in Layer 1 payload
  - pdcch-ConfigSIB1
    - 4 bit for CORESET0 index
    - 4 bit for searchSpace0 index
    - Use for SIB1 acquisition

message mib (Period of 80ms):

```
{
  systemFrameNumber '110011'B,
  subCarrierSpacingCommon scs30or120,
  ssb-SubcarrierOffset 6,
  dmrs-TypeA-Position pos3,
  pdcch-ConfigSIB1
  {
    controlResourceSetZero 0,
    searchSpaceZero 0
  },
  cellBarred notBarred,
  intraFreqReselection allowed,
  spare '0'B
}
```

2. Cell Search

System Information Block 1 (Period of 160ms)

cellSelectionInfo	RSRP, RSRQ information for cell (re)selection
cellAccessRelatedInfo	List of PLMN-IdentityInfo: PLMN ID, NR Cell ID, tac,…
connEstFailureControl	Decide behavior after UE experience repeated RRC establishment failures
si-SchedulingInfo	Information for other System information broadcast (SIB2 to SIB 9)
servingCellConfigCommon	DL and UL common config (frequency band, SCS, bandwidth part, bcch, pcch, rach, pdcch, pucch, pusch, pdsch,…)
ue-TimerAndConstants	t300, t301, t310, n310, t311, n311, t319

3. Random Access Procedure

Why?

- Achieve UL synchronization between UE and gNB
- Obtain the resource for Message 3 (e.g, RRC Setup Request)

When?

- Initial access from RRC_IDLE;
- RRC Connection Re-establishment procedure;
- Handover;
- DL or UL data arrival when UE is Out-of-Sync;
- Transition from RRC_INACTIVE;
- Request for Other SI;
- Beam failure recovery;
- …

3. Random Access Procedure

Two type of RACH:

- Contension Based Random Access (CBRA)
  - UE select a random preamble at the risk of experiencing contension
  - Used for all procedures above
- Contension Free Random Access (CFRA)
  - UE use a dedicated preamble provided by RRC signaling/Layer 1 signaling
  - Used when
    - Handover
    - DL data arrival when UE is Out-of-Sync
    - Request of System Information
    - Beam Failure Recovery
    - Synchronous Reconfiguration
    - ENDC SCell Addition

3. Random Access Procedure

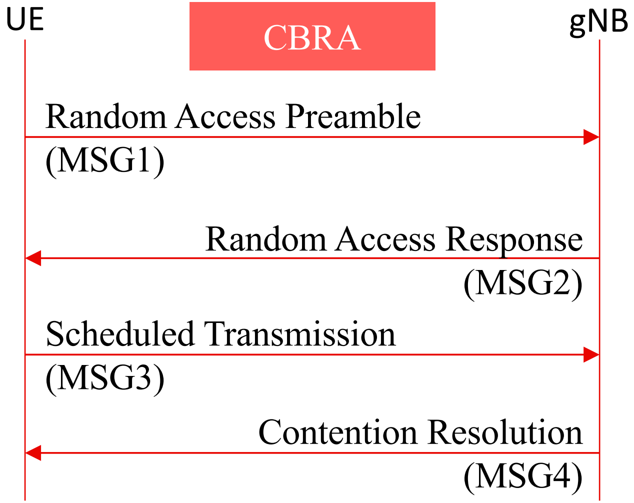
Contention Based Random Access (CBRA):

B1. Transmitting MSG1 preamble on PRACH

- Contention if multiple UE select the same PRACH occasion and PRACH preamble

B2. UE receives a Random Access Response

- RAPID
  - corresponding to PRACH preamble
- Timing Advance Command
  - synchronise the UL transmission timing at gNB
- Uplink Grant
  - UL resource allocation for MSG3
- Temporary C-RNTI
  - If RACH successfully complete, it becomes the allocated C-RNTI



3. Random Access Procedure

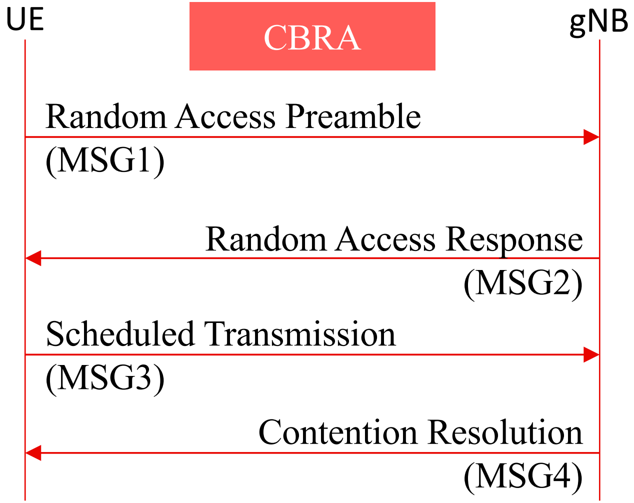
Contension Based Random Access (CBRA):

B3. MSG3 includes:

- CCCH msg
- DCCH/DTCH msg + C-RNTI

B4. MSG4 + “UE Contention Resolution ID” (MAC CE or C-RNTI)

- UE decode successfully and “UE ID” is Correct: The procedure is successful & UE send back an HARQ ACK
- UE decode successfully and “UE ID” is InCorrect: The procedure is unsuccessful & UE restarts the procedure
- UE decode unsuccessfully: If the Contention Resolution timer expire, the procedure is restarted

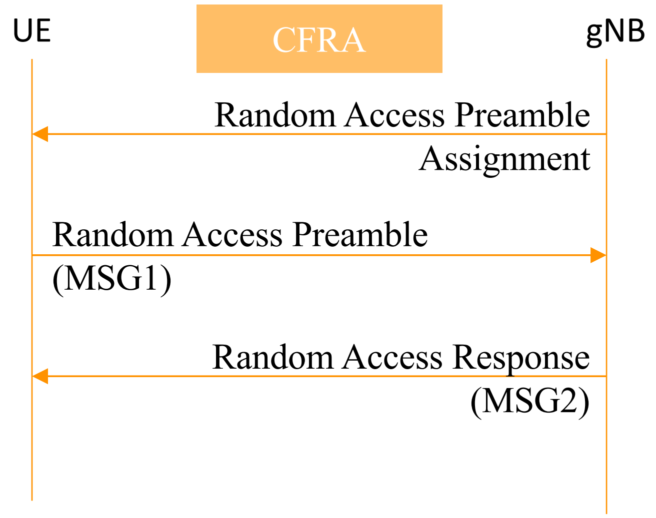


3. Random Access Procedure

Contension Free Random Access (CFRA):

- The difference from CBRA
    - RACH Preamble Assigment done by RRC Configuration

  → No Contention Resolution



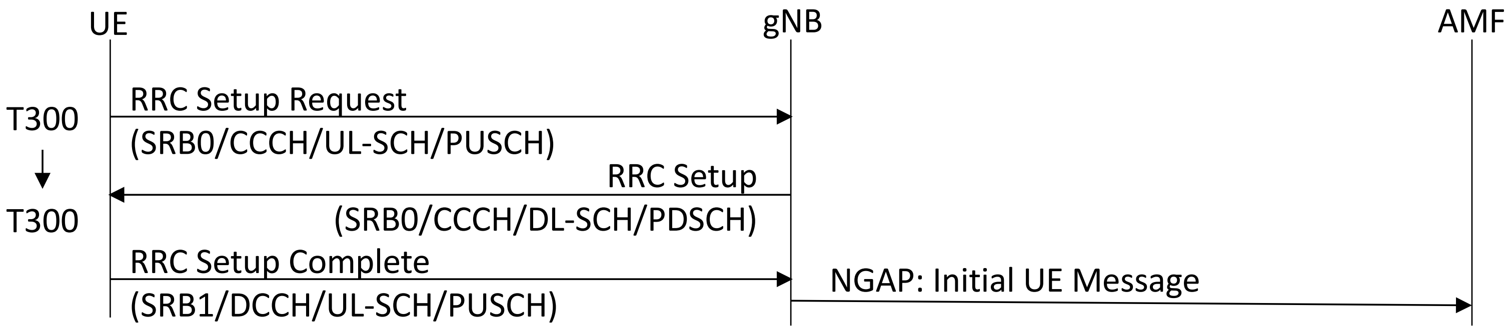
4. RRC Connection Establishment

Purpose:

- Establish an RRC connection
- Transfer the initial NAS dedicated message from the UE to the network

Triggering Source:

- Network (e.g. Paging message)
- UE (e.g. end-user try to browse the internet or to send an email)



4. RRC Connection Establishment

RRCSetupRequest <3GPP TS 38.331>		
ue-Identity	CHOICE	
	ng-5G-S-TMSI-Part1	randomValue
	BIT STRING {39 bits}	BIT STRING {39 bits}
establishmentCause	emergency, highPriorityAccess, mt-Access, mo-Signalling, mo-Data, mo-VoiceCall, mo-VideoCall, mo-SMS, mps-PriorityAccess, mcs-PriorityAccess	

Message content:

- UE Identity: 39 bits
- Establishment cause: 4 bits
- Spare: 1 bit
- Choice between 2 UE ID types: 1 bit
- Choice between 4 CCCH messages : 2 bit + 1 bit for message extensions

4. RRC Connection Establishment

Upon receiving RRCSetup message:

- UE stops the T300 timer
- UE make the transition to RRC Connected mode & transmit RRCSetupComplete

RRCSetup is divided into 2 main section:

- Radio Bearer configuration
  - SRB identity to be setup: 1
  - PDCP layer Config(Optional)
- Master Cell Group configuration
  - RLC layer Config
  - MAC layer Config
  - Physical layer Config
  - spCell Config

4. RRC Connection Establishment

RRCSetupComplete <3GPP TS38.331>			
selectedPLMN-Identity	1 to 12		
registeredAMF	plmn-Identity	mcc	3 digits
		mnc	2 or 3 digits
	amf-Identifier	BIT STRING {24 bits}	
guami-Type	native, mapped		
s-NSSAI-List	SEQUENCE {1 to 8 instances}		
	S-NSSAI	sst	BIT STRING {8 bits}
		sst-SD	BIT STRING {32 bits}
dedicatedNAS-Message	OCTET STRING		
ng-5G-S-TMSI-Value	CHOICE		
	ng-5G-S-TMSI	ng-5G-S-TMSI-Part2	
	BIT STRING {48 bits}	BIT STRING {9 bits}	

```
message c1 : rrcSetupComplete :
  {
    rrc-TransactionIdentifier 0,
    criticalExtensions rrcSetupComplete :
    {
      selectedPLMN-Identity 1,
      registeredAMF
      {
        amf-Identifier '00111110 00011001 00001010'B
      },
      guami-Type native,
      dedicatedNAS-Message
'7E004172000BF254F2403E190AD981DD182E04F070F070'H
    }
  }
```

Registration Request to forward to AMF

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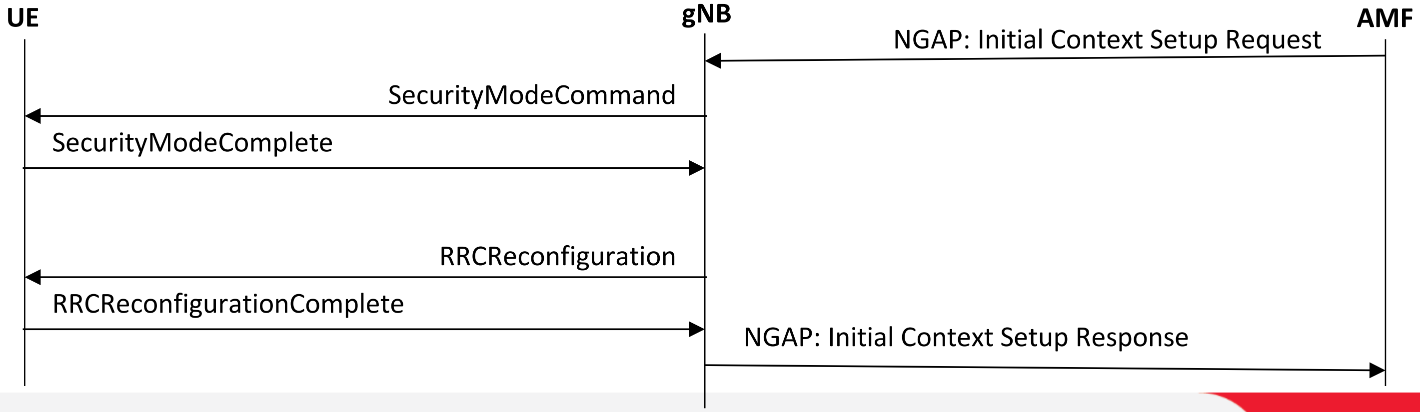
5. Initial Context Setup

After UE has established an RRC Connection (thus configured SRB1) and completed the NAS Authentication and Security, the Initial Context Setup procedure can be used to:

- Setup UE Context at the gNB
- Activate ciphering and integrity protection with gNB
- Configure SRB 2
- Configure one or more DRB



5. Initial Context Setup

The AMF initiates the procedure by sending NGAP Initial Context Setup Request message

- AMF UE NGAP ID/RAN UE NGAP ID
  - Address the UE Context
- NAS PDU: Registration Accept
- PDU Session Resource setup request List
  - If included, gNB will behave the same as in PDU Session Resource Setup procedure to establish PDU Session
    - PDU Session IDs
    - UL TEIDs

Initial Context Setup Request	
Information Elements	**Presence**
AMF UE NGAP ID	M
RAN UE NGAP ID	M
Old AMF	O
UE Aggregate Maximum Bit Rate	C
Core Network Assistance Information	O
Global Unique AMF ID	M
PDU Session Resource Setup Request List	O
Allowed NSSAI	M
UE Security Capabilities	M
Security Key	M
Trace Activation	O
Mobility Restriction List	O
UE Radio Capability	O
Index to RAT/Frequency Selection Priority (RFSP)	O
Masked IMEISV	O
NAS PDU	O
Emergency Fallback Indicator	O
RRC Inactive Transition Report Request	O
UE Radio Capability for Paging	O
Redirection for Voice EPS Fallback	O

5. Initial Context Setup

The gNB start Integrity protection & Ciphering process for Downlink and Uplink stream by sending Security Mode Command message

- Ciphering Algorithm: nea0, nea1, nea2, nea3
- Integrity Algorithm: nia0, nia1, nia2, nia3

The UE acknowledges with Security Mode Command message

```
message c1 : securityModeCommand :
    {
     rrc-TransactionIdentifier 0,
     criticalExtensions securityModeCommand :
       {
        securityConfigSMC
        {
          securityAlgorithmConfig
          {
            cipheringAlgorithm nea0,
            integrityProtAlgorithm nia2
          }
        }
       }
    }
```

```
message c1 : securityModeComplete :
    {
     rrc-TransactionIdentifier 0,
     criticalExtensions securityModeComplete :
       {
       }
    }
```

5. Initial Context Setup

RRC Reconfiguration message ↔ RRC Reconfiguration Complete message

- Configure SRB2
- Configure a set of DRB
- Configure Measurement Event
- Master/SecondaryCellGroup Config

```
message c1 : rrcReconfiguration :

    rrc-TransactionIdentifier 0,
    criticalExtensions rrcReconfiguration :
     {
       radioBearerConfig
       {
         srb-ToAddModList
         {
           {
             srb-Identity 2
           }
         },
```

```
drb-ToAddModList
      {
        {
          cnAssociation sdap-Config :
          {
            pdu-Session 1,
            sdap-HeaderDL absent,
            sdap-HeaderUL present,
            defaultDRB TRUE,
            mappedQoS-FlowsToAdd
            {
              6
            }
          },
          drb-Identity 1,
          pdcp-Config
          {
            drb
            {
              discardTimer infinity,
              pdcp-SN-SizeUL len18bits,
              pdcp-SN-SizeDL len18bits,
              headerCompression notUsed : NULL,
              statusReportRequired true
            },
            t-Reordering ms3000
          }
```

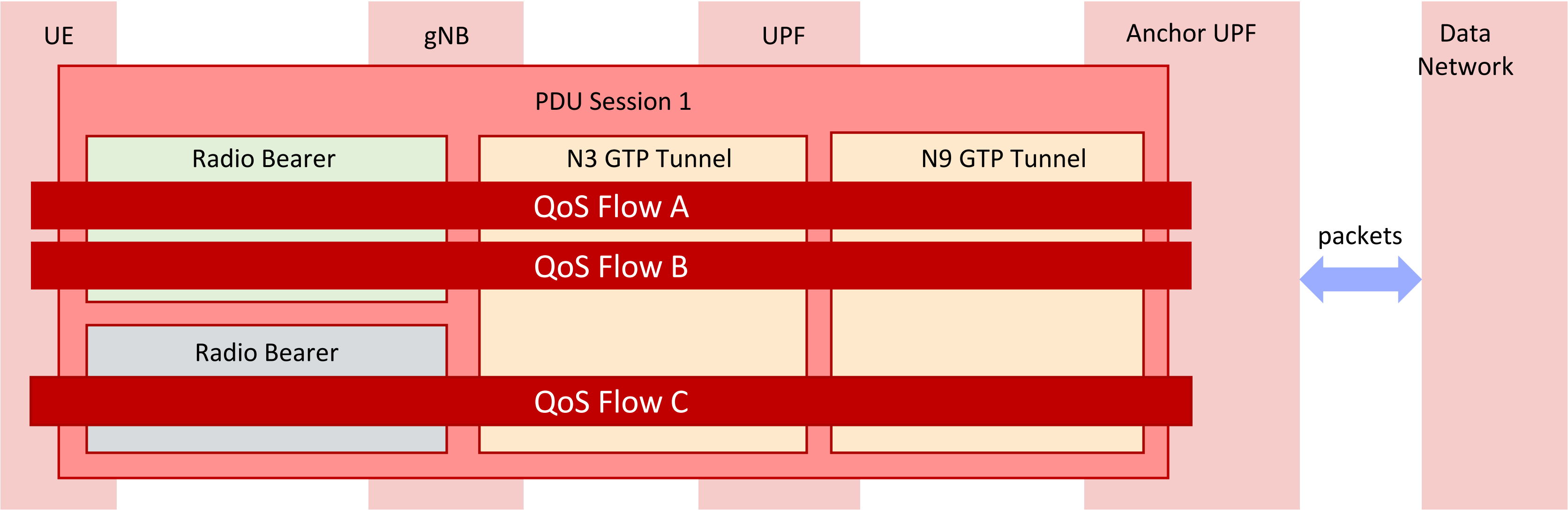
5. Initial Context Setup

The gNB completes the Procedure with INITIAL CONTEXT SETUP RESPONSE message to the AMF

- If INITIAL CONTEXT SETUP REQUEST includes PDU Session Setup Request
    - PDU Session Resource Setup Response List
        - Signal the successful setup of PDU Sessions
        - DL TEIDs
    - PDU Session Resource Failed to Setup List
        - Cause

6. PDU Session Establishment

PDU Session Establishment is the process of establishing a User plane Connectivity between the UE and the 5G Core Network.



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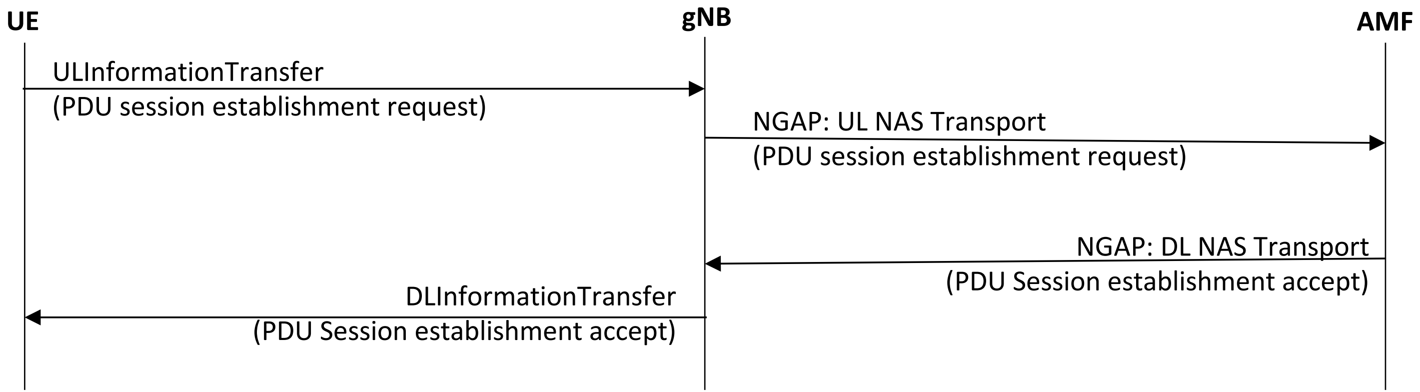
6. PDU Session Establishment

The procedure flow:

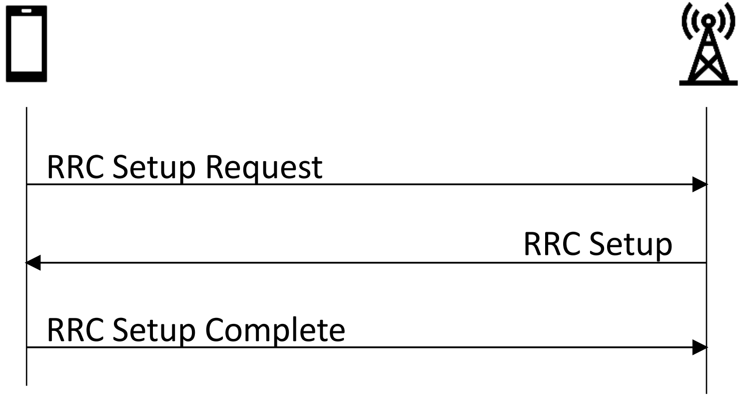


- The AMF modifies the Session Management Context based on the message from the gNB
- SMF signals PDU session updates to the UPF
- UE & Network can now transfer data

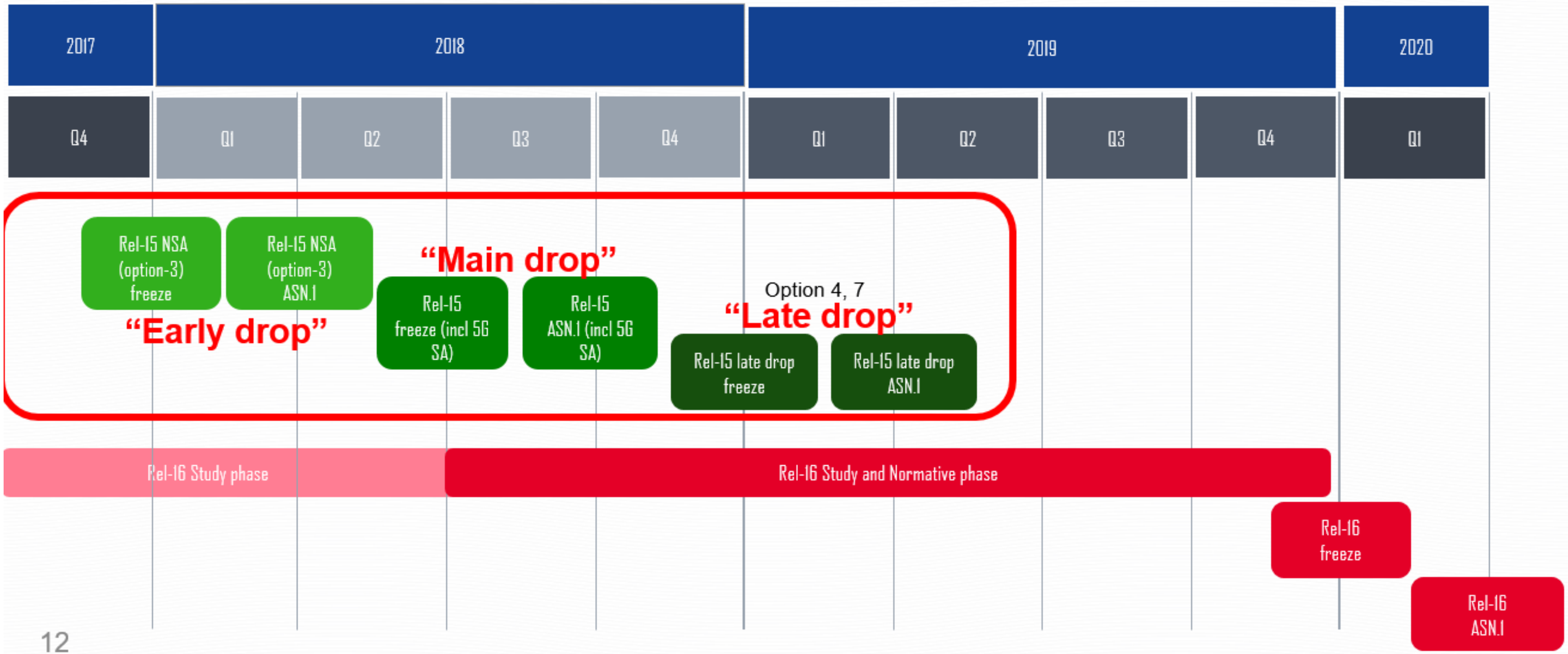
7. Simulation

Simulate the RRC Conntection Establishment

- Features
  - KPI counter
    - $\text{Success rate} = \frac{\mathit{RRC\ Setup\ Complete}}{\mathit{RRC\ Setup\ Request}}$
  - Psuedo-random drop rate
  - Multiple UE



3GPP release



8. 3GPP Release

TS 38.104	NR; Base Station (BS) radio transmission and reception

TS 38.201	NR; Physical layer; General description
TS 38.202	NR; Services provided by the physical layer
TS 38.211	NR; Physical channels and modulation
TS 38.212	NR; Multiplexing and channel coding
TS 38.213	NR; Physical layer procedures for control
TS 38.214	NR; Physical layer procedures for data
TS 38.215	NR; Physical layer measurements
TS 38.321	NR; Medium Access Control (MAC) protocol specification
TS 38.322	NR; Radio Link Control (RLC) protocol specification
TS 38.323	NR; Packet Data Convergence Protocol (PDCP) specification
TS 38.331	NR; Radio Resource Control (RRC); Protocol specification
TS 38.340	NR; Adaptation layer (SDAP)

TS 38.141-1	NR; Base Station (BS) conformance testing Part 1: Conducted conformance testing
TS 38.141-2	NR; Base Station (BS) conformance testing Part 2: Radiated conformance testing
TR 38.810	NR; Study on test methods

TS 38.133	NR; Requirements for support of radio resource management

TS 38.401	NG-RAN; Architecture description
TS 38.410	NG-RAN; NG general aspects and principles
TS 38.411	NG-RAN; NG layer 1
TS 38.412	NG-RAN; NG signalling transport
TS 38.413	NG-RAN; NG Application Protocol (NGAP)
TS 38.414	NG-RAN; NG data transport
TS 38.415	NG-RAN; PDU Session User Plane protocol
TS 38.420	NG-RAN; Xn general aspects and principles
TS 38.421	NG-RAN; Xn layer 1
TS 38.422	NG-RAN; Xn signalling transport
TS 38.423	NG-RAN; Xn Application Protocol (XnAP)
TS 38.424	NG-RAN; Xn data transport
TS 38.425	NG-RAN; NR user plane protocol
TS 38.455	NG-RAN; NR Positioning Protocol A (NRPPa)
TS 38.460	NG-RAN; E1 general aspects and principles
TS 38.461	NG-RAN; E1 layer 1
TS 38.462	NG-RAN; E1 signalling transport
TS 38.463	NG-RAN; E1 Application Protocol (E1AP)
TS 38.470	NG-RAN; F1 general aspects and principles
TS 38.471	NG-RAN; F1 layer 1
TS 38.472	NG-RAN; F1 signalling transport
TS 38.473	NG-RAN; F1 Application Protocol (F1AP)
TS 38.474	NG-RAN; F1 data transport
TS 38.475	NG-RAN; F1 interface user plane protocol